

# Lecture 17: Business activity, unemployment and inflation



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and Society

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"The economy is a system of interconnected parts, and investment is the spark that drives its growth."



# Lecture overview

- Simple Keynesian Model of Business Activity
- Alternative Perspectives on Aggregate Supply
- Output, Unemployment and Inflation
- Inflation Targeting and Unemployment

# Simple Keynesian Model of Business Activity

# Simple Keynesian Model

- **Key macroeconomic variables:** GDP and national income are often used interchangeably.
- **Circular flow of income model:** Illustrates income flows between different sectors of the economy and their impact on aggregate demand.
- **Fluctuations in aggregate demand:** Economists believe that these fluctuations drive the business cycle.
- **Keynesian model development:** Revisits the circular flow model and focuses on the determination of national income (GDP) and the effect of aggregate demand changes.
- **Assumption:** Prices are constant in the initial model; later, effects on output and prices are considered.

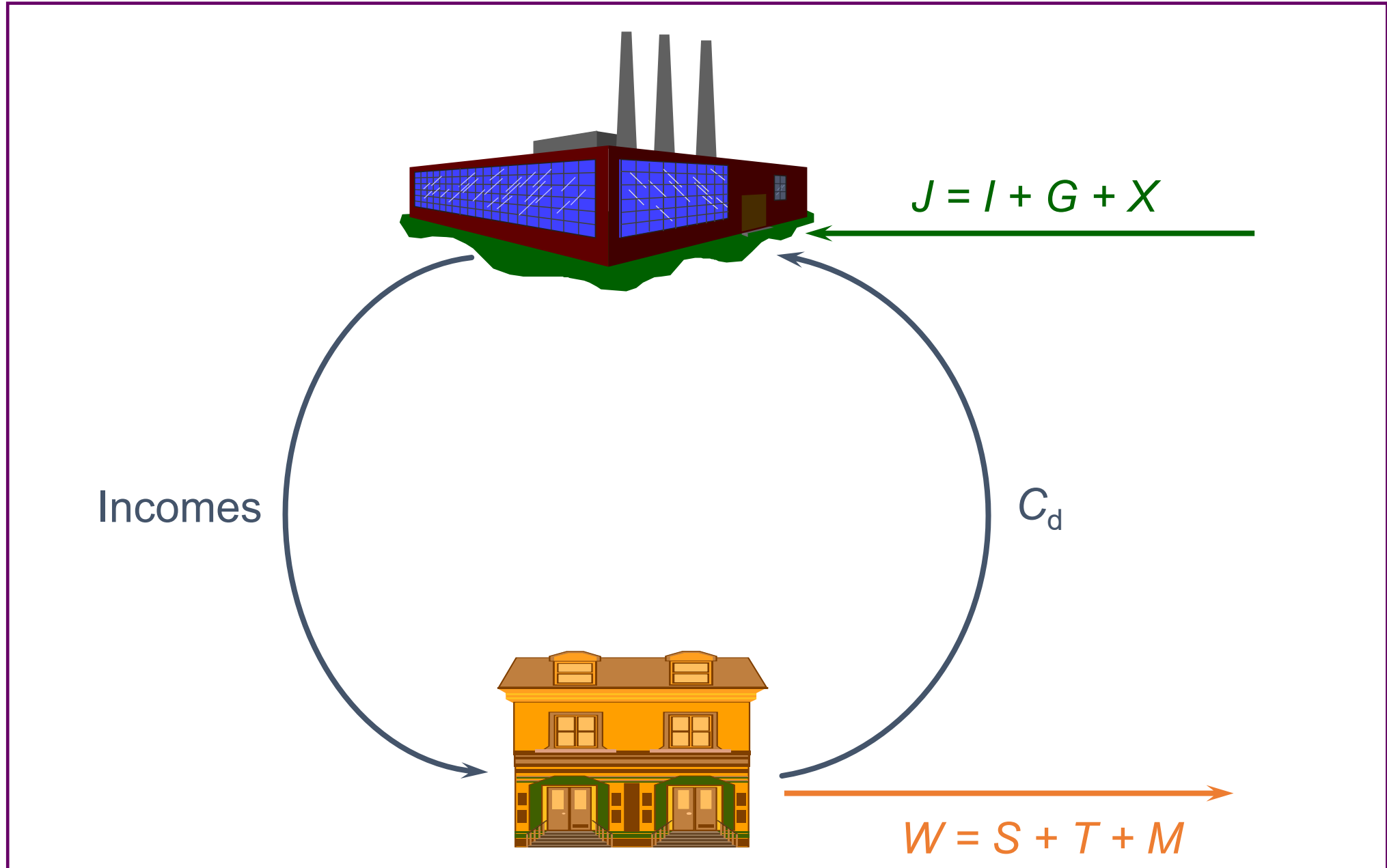
# Simple Keynesian Model

- The importance of aggregate demand
  - effects of  $J > W$
  - effects of  $W > J$

# Impact of Changes in Aggregate Demand

- **Initial equilibrium:** Injections equal withdrawals.
- **Increase in injections** (e.g., firms investing more): Leads to higher aggregate demand, more income, and employment due to the multiplier effect.
- **Multiplier effect:** Increased income leads to more consumption, which further increases demand and employment, but gradually slows as withdrawals (savings, taxes, imports) rise.
- **Fall in injections:** Causes a multiplied fall in national income and employment.

# The circular flow of income

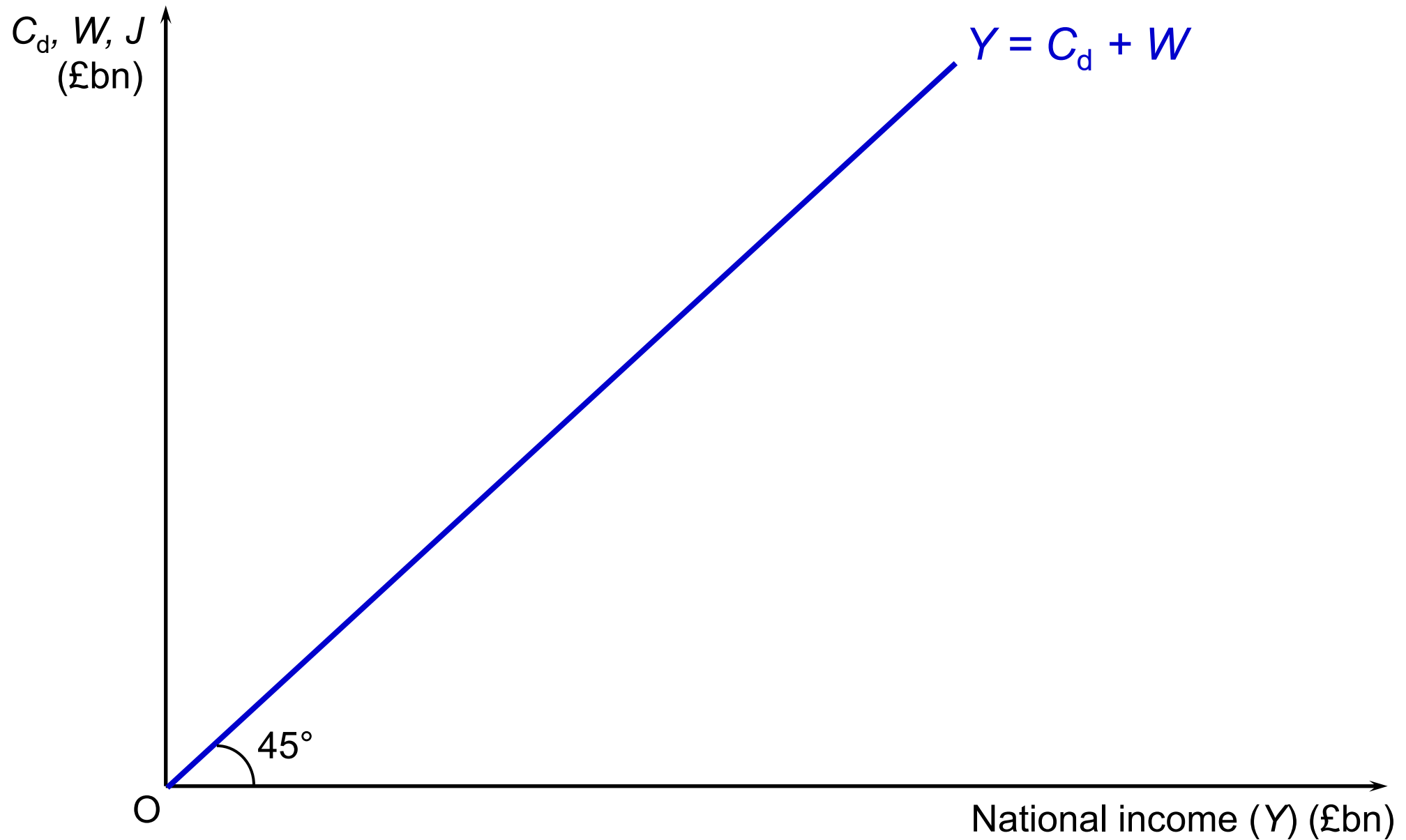


# Simple Keynesian Model

- The importance of aggregate demand
  - effects of  $J > W$
  - $J > W \longrightarrow Y \uparrow \longrightarrow W \uparrow \text{ until } J=W$
  - effects of  $W > J$
  - $J > W \longrightarrow Y \downarrow \longrightarrow W \downarrow \text{ until } J=W$
- **Keynesian Cross:** Aggregate demand determines equilibrium national income and is plotted against national income on a 45-degree diagram.



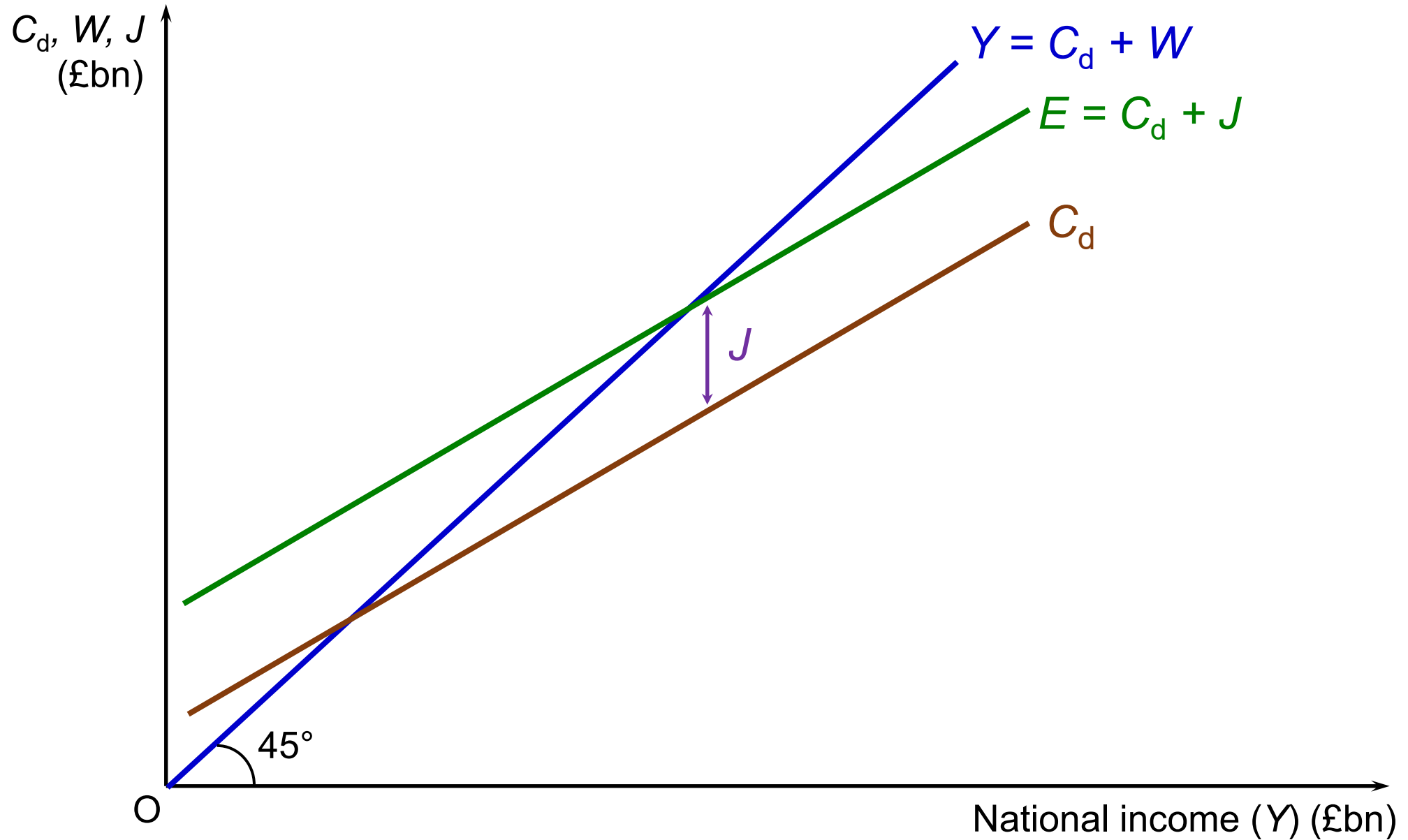
# Equilibrium national income



# Keynesian 45° Line Diagram

- **45° line:** Represents where national income ( $Y$ ) equals consumption ( $C_d$ ) + withdrawals ( $W$ ).
- **Aggregate expenditure line (E):** Plots total spending, including consumption and injections.
- **Consumption line ( $C_d$ ):** Flatter than the 45° line because not all of the increase in GDP is spent domestically.
- **E line:** Shifted upward from the  $C_d$  line by the amount of injections

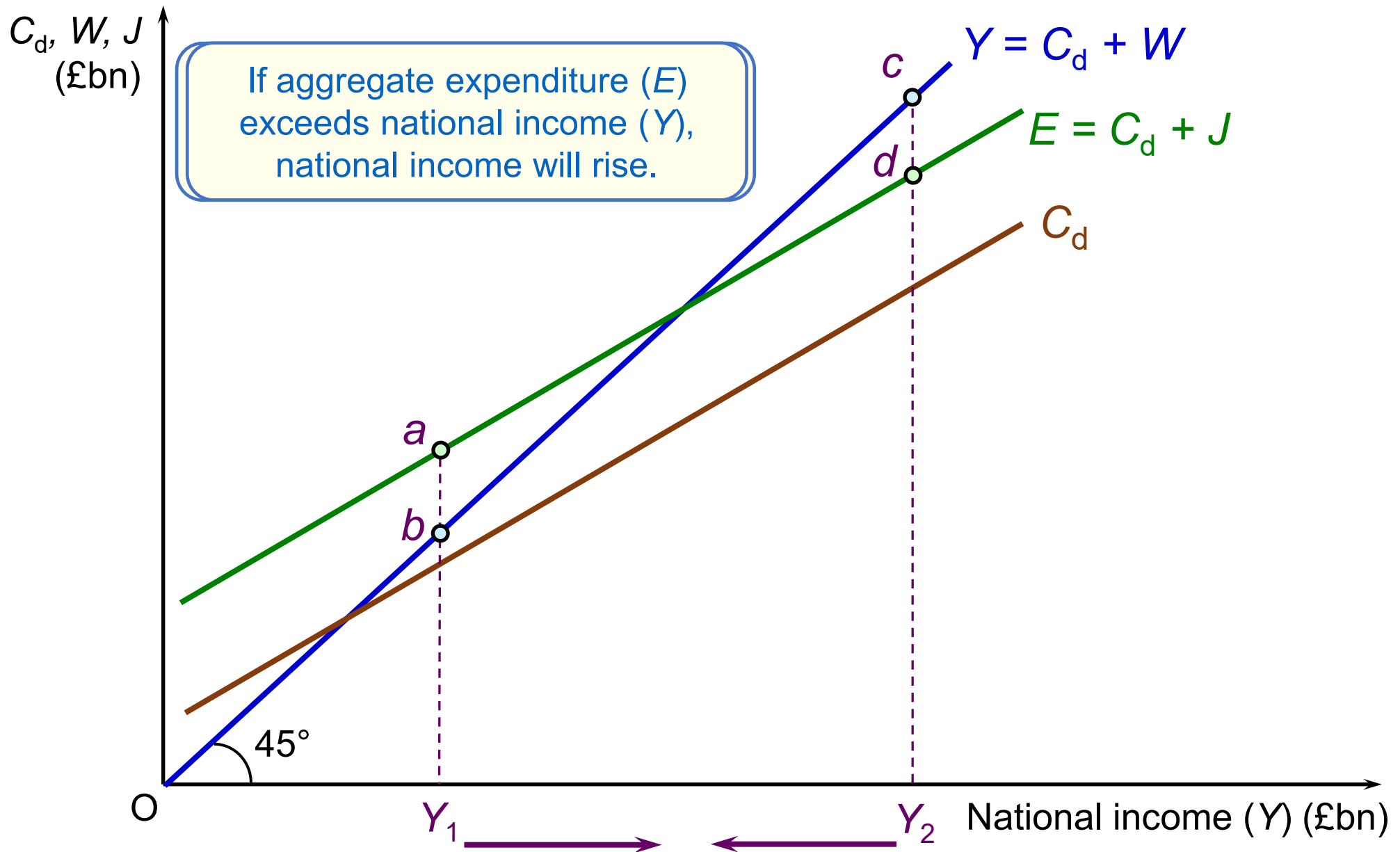
# Equilibrium national income



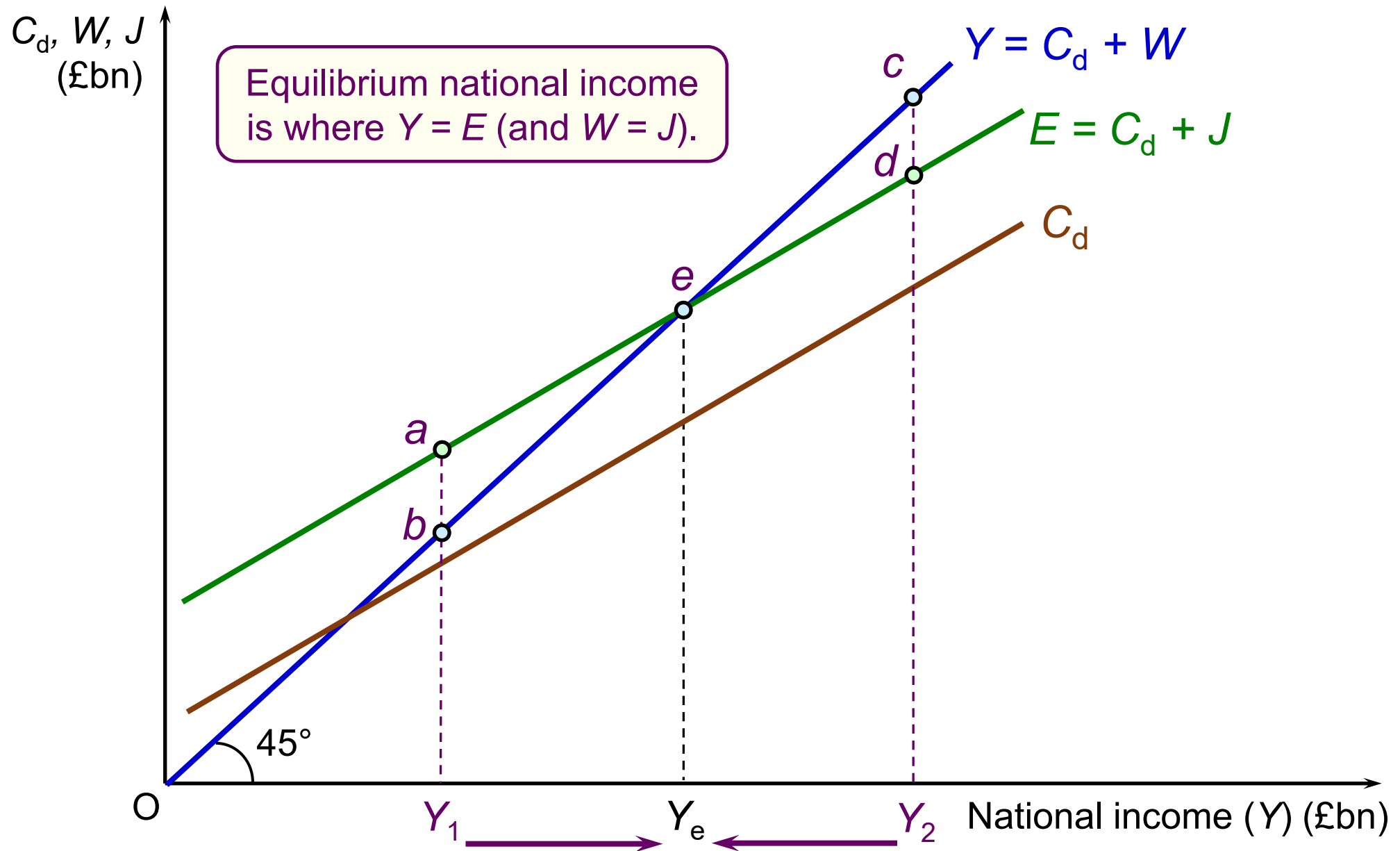
# Excess Demand and Insufficient Demand

- effects of  $E > Y$ : national income rises
- effects of  $Y > E$ : national income falls
- **Excess demand:** If aggregate expenditure exceeds national income, firms increase production and employment, raising GDP.
- **Insufficient demand:** If GDP exceeds aggregate expenditure, firms reduce production and employment, causing GDP to fall until equilibrium is restored.
- **Equilibrium:** Occurs when national income equals aggregate expenditure ( $Y = E$ ).

# Equilibrium national income



# Equilibrium national income



Q Given the following data (in £bn), what will happen to equilibrium national income?

$$C = 100, I = 20, M = 30, X = 25, S = 18, T = 28, G = 26$$

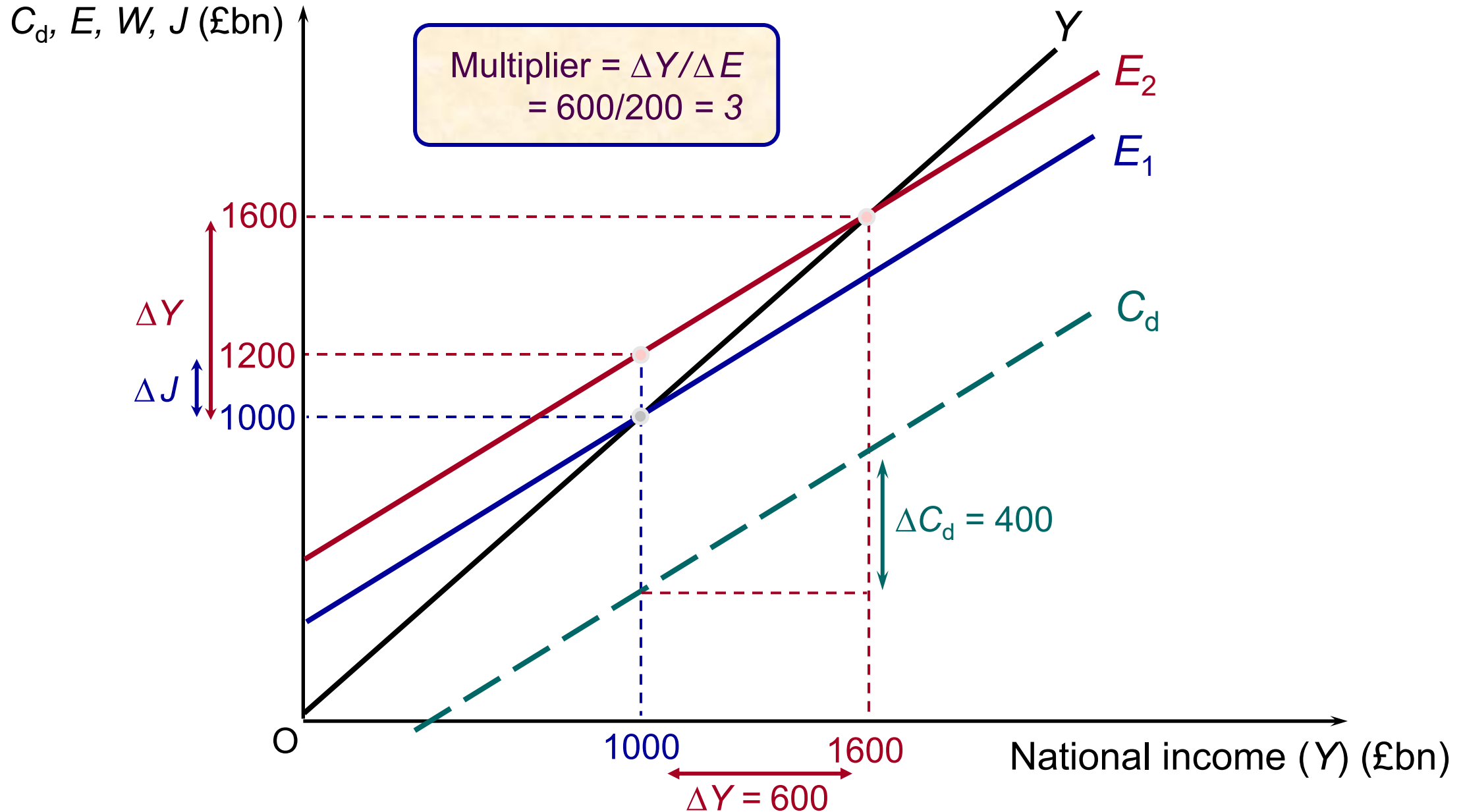
- A. Increase
- B. Decrease
- C. Stay the same

# The Multiplier Effect

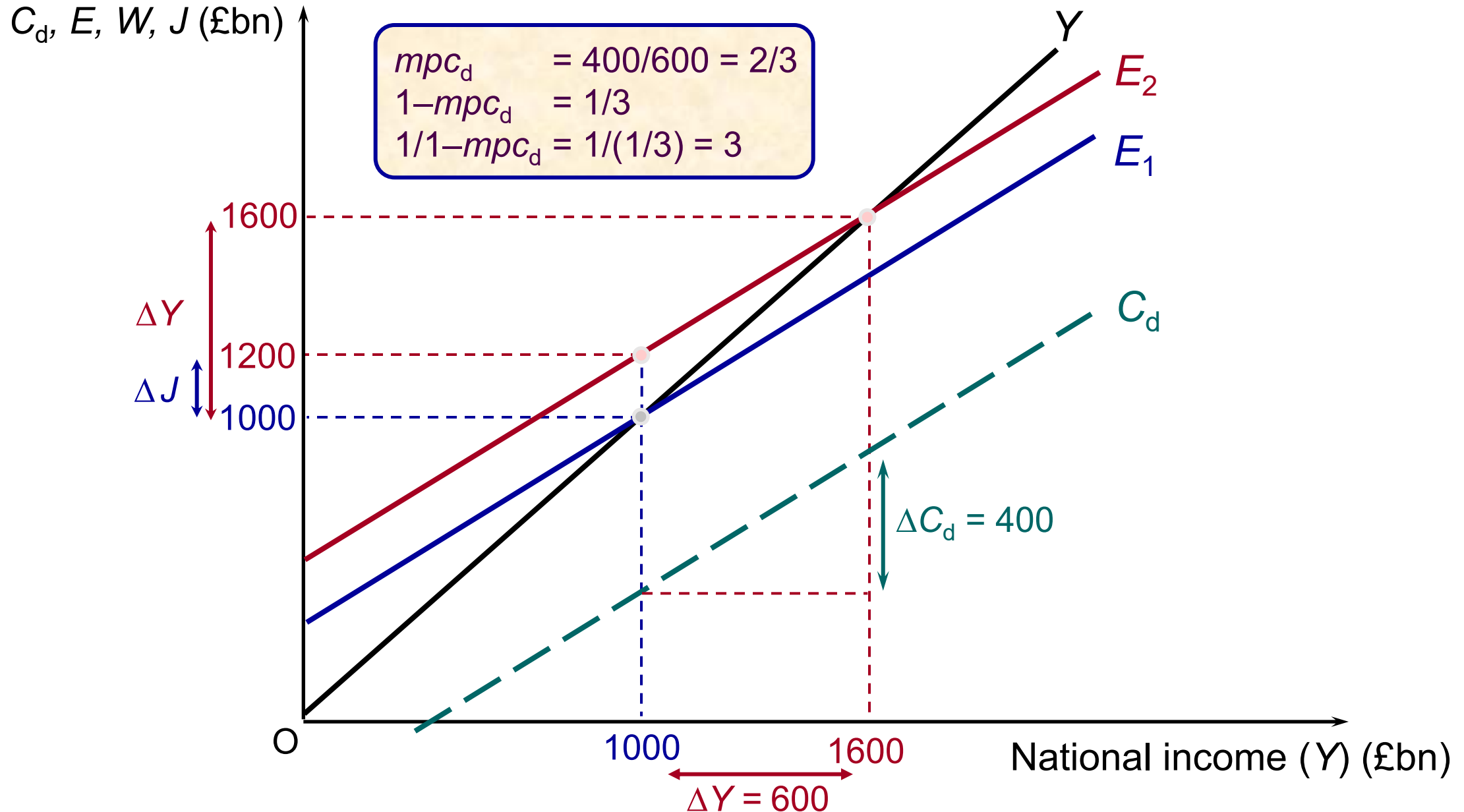
- The multiplier: The rise in GDP is larger than the rise in aggregate expenditure
  - definition of the multiplier ( $k$ )
    - $k = \Delta Y / \Delta E$
  - **Example:** A £10 billion rise in aggregate expenditure could lead to a £30 billion rise in GDP, with a multiplier of 3.
  - **Marginal Propensity to Consume (MPC):** Determines the size of the multiplier, with higher MPC leading to a larger multiplier
    - $mpc_d = \Delta C_d / \Delta Y$
    - Multiplier ( $k$ ) =  $1 / (1 - mpc)$



# The multiplier: a rise in aggregate expenditure



# The multiplier: a rise in aggregate expenditure



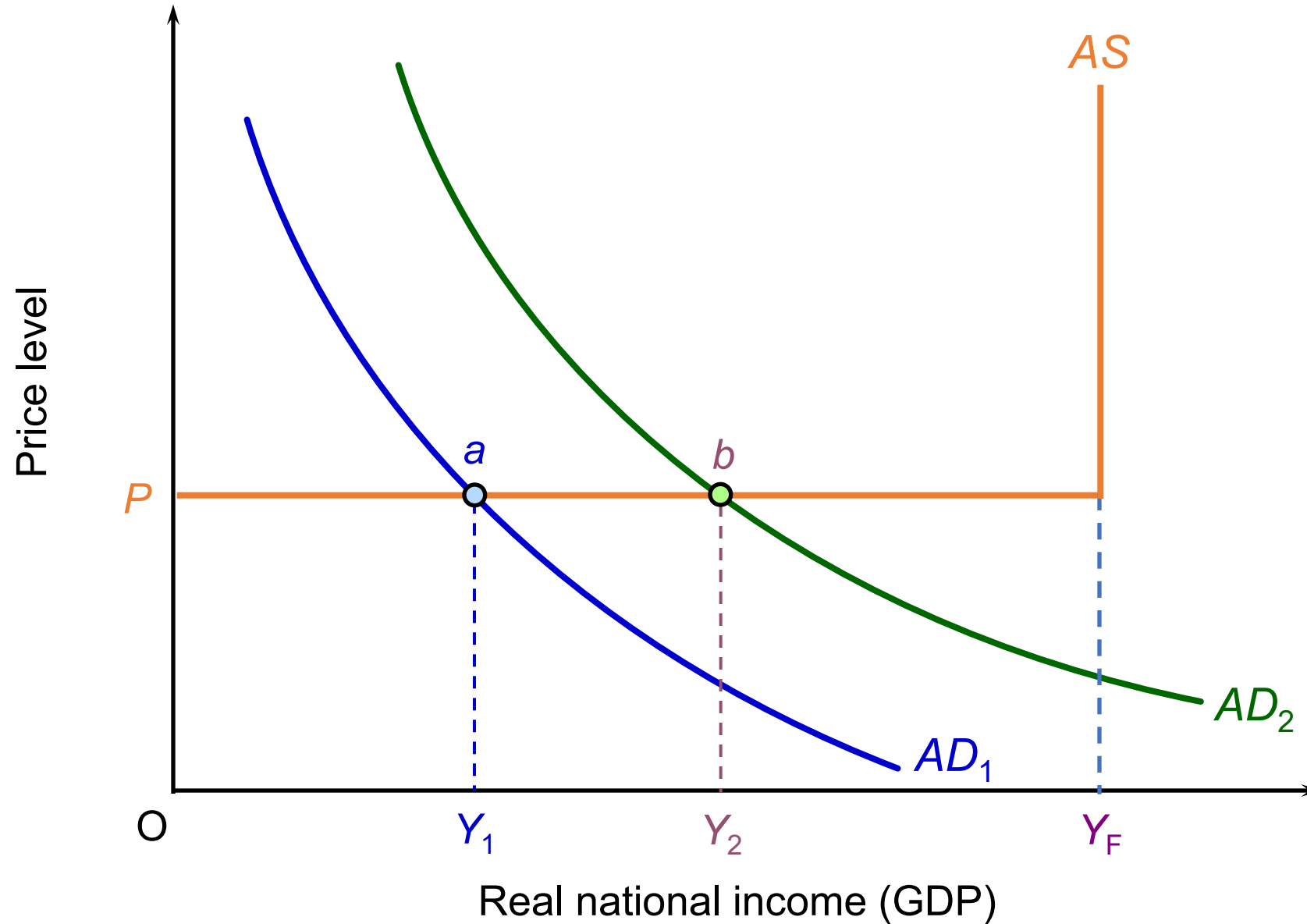
If a rise in national income of £100m causes consumption of domestic goods and services to rise by £60m, the multiplier is:

- A. 6
- B. 4
- C. 2.5
- D. 2
- E. 1.67

# Alternative Perspectives on Aggregate Supply

- Short-run aggregate supply curve (*SRAS*)
  - flexibility of aggregate supply shows how output responds to a change in aggregate demand
  - various perspectives on aggregate supply
  - extreme Keynesian position
    - *SRAS* horizontal up to  $Y_F$ 
      - changes in *AD* purely reflected in real output ( $Y$ )
    - *SRAS* vertical at  $Y_F$

## Different short-run aggregate supply curves

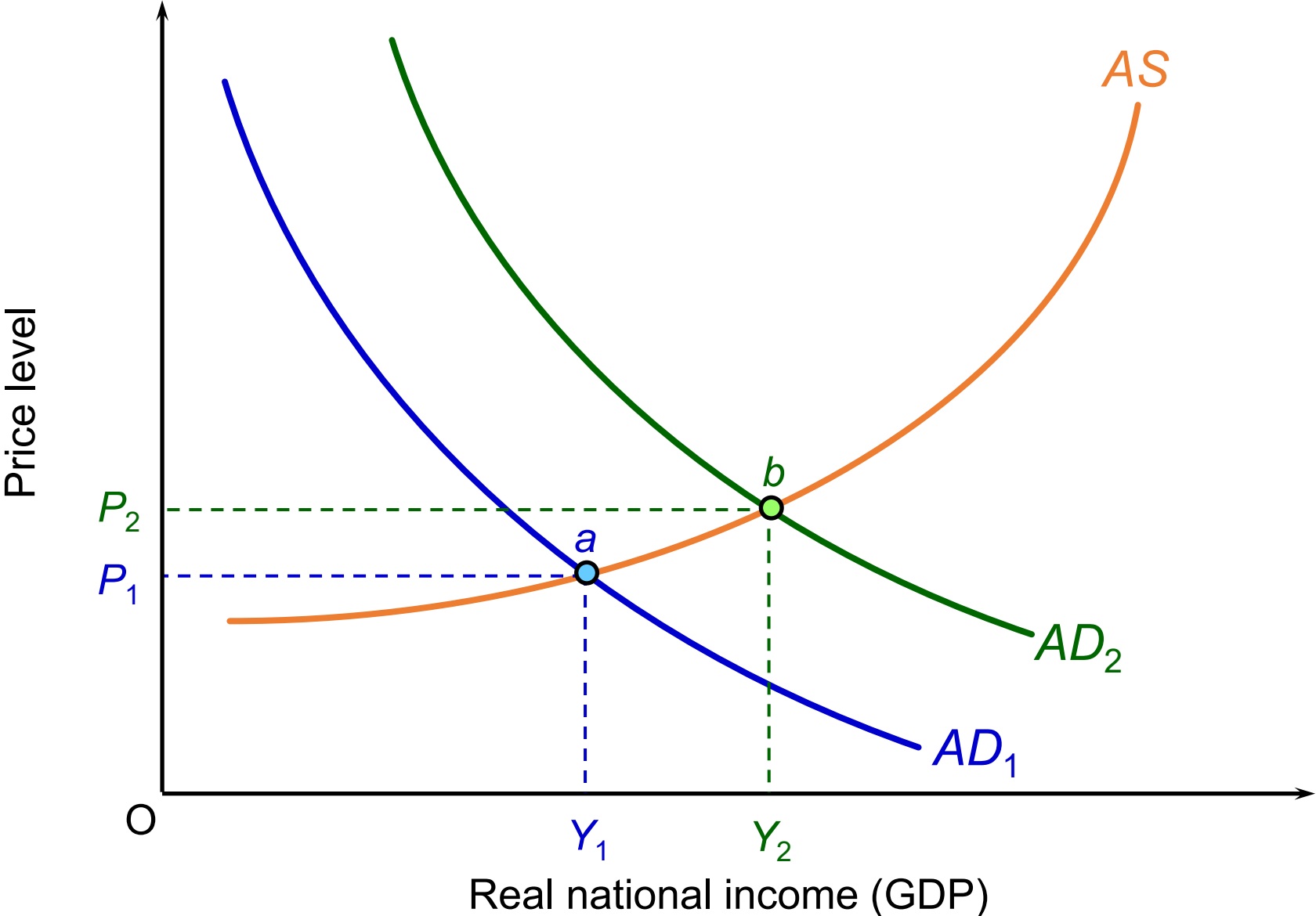


Aggregate supply totally dependent on aggregate demand up to full-employment income

**(a)** The extreme Keynesian position

- Short-run aggregate supply curve (*SRAS*) (cont.)
  - moderate (mainstream) position
    - *SRAS* upward sloping: gets steeper as it approaches  $Y_F$
    - some stickiness in wages and prices
    - diminishing returns

# Different short-run aggregate supply curves



Aggregate supply becoming less and less responsive to aggregate demand as full employment is reached

(b) Mainstream position



# The New Classical Position



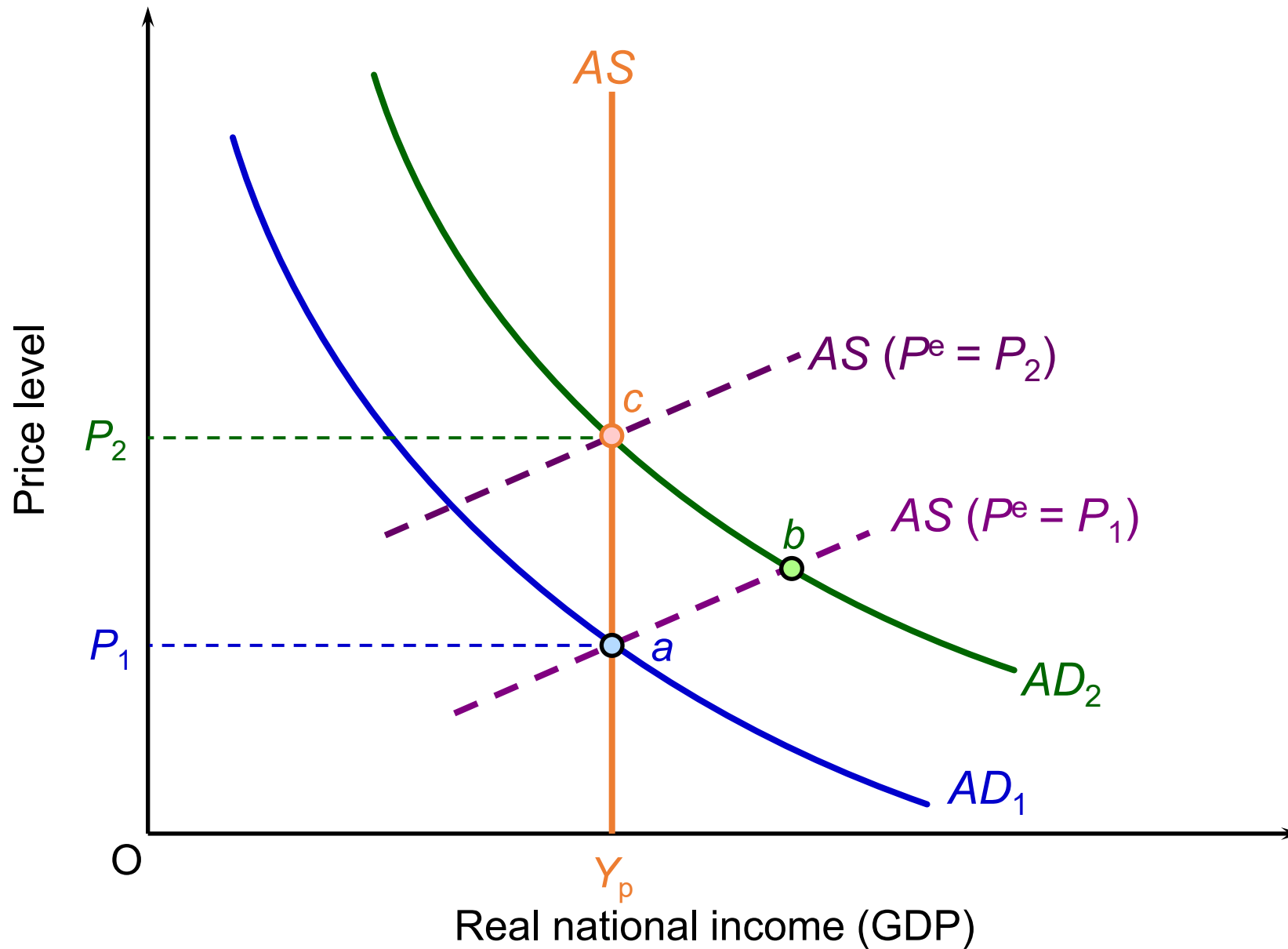
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- New classicists argue that the Short-Run Aggregate Supply (SRAS) curve is vertical at potential output ( $Y_P$ ). Assumptions:
  - Continuous market clearing: all markets adjust to equilibrium continuously
  - Rational expectations: people use all available information to predict macroeconomic variables accurately.

# The New Classical Position

- Anticipated changes in aggregate demand lead to price increases but no real change in national income, which remains at potential output (YP).
- Unanticipated changes in aggregate demand lead to temporary deviations from YP before the economy adjusts back.
- Supply-side economics: Governments should focus on supply (encouraging competition and enterprise) rather than just demand.
- Real business cycles: Fluctuations in aggregate supply (e.g., technological shocks, regulatory changes) can affect potential output and contribute to economic volatility.

## Different short-run aggregate supply curves



Aggregate supply is independent of aggregate demand when changes in  $AD$  are anticipated

(c) New classical position

# Output, Unemployment and Inflation

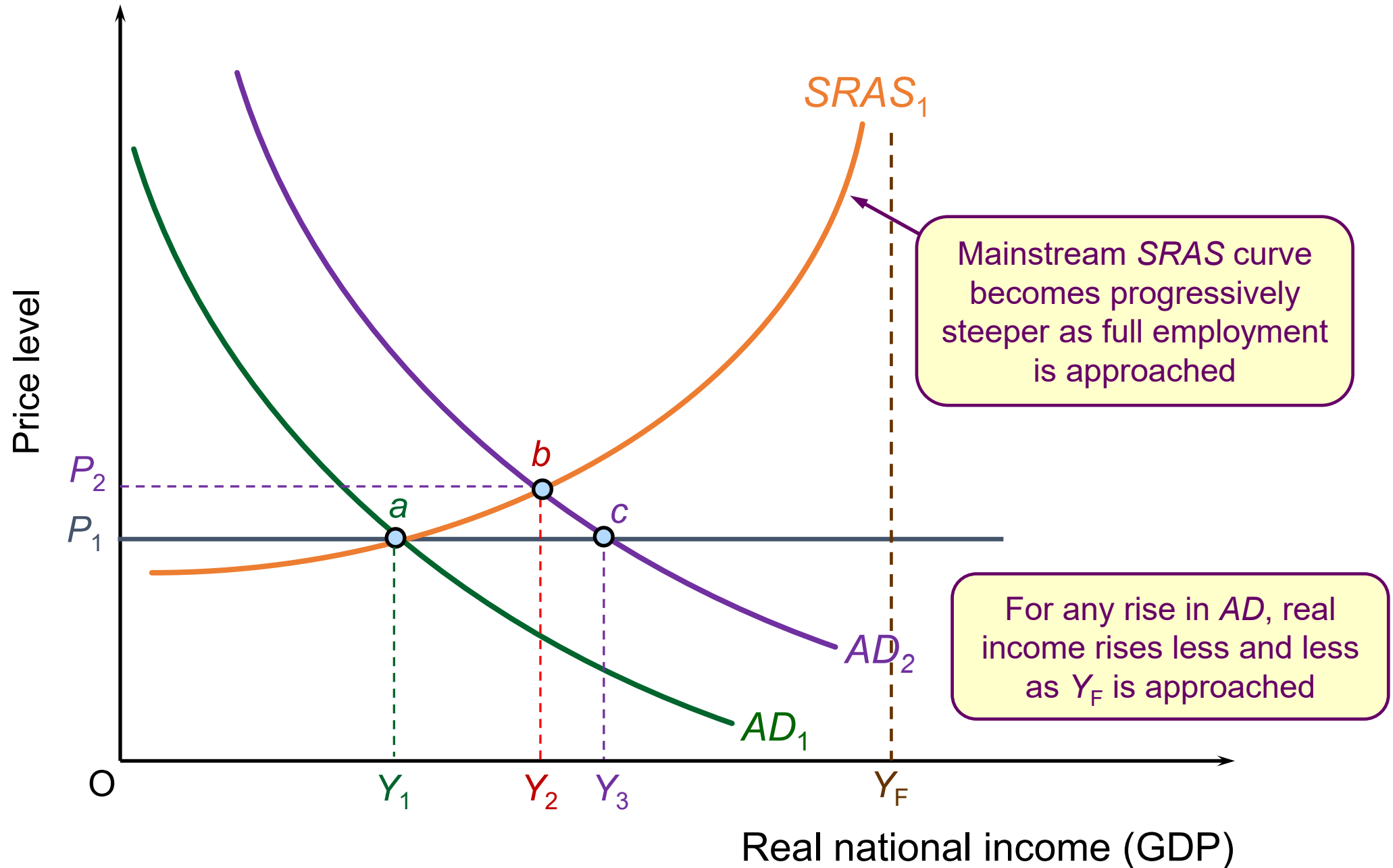
# Output, Unemployment and Inflation



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- Business cycles arise from fluctuations in aggregate demand and aggregate supply, affecting output, unemployment, and inflation.
- The AD/AS model helps analyze these relationships.
- A rise in aggregate demand can reduce unemployment and increase prices, especially before full employment is reached.
- As national output rises, inflationary pressures become more significant, and the multiplier effect becomes smaller.
- At higher output levels, a larger portion of demand increases results in higher prices rather than more output.

## Fluctuations in aggregate demand and the size of the multiplier



# Other Types of Inflation and Unemployment

- Not all inflation and unemployment are due to aggregate demand.
  - Frictional and structural unemployment are not linked to aggregate demand.
  - Cost-push and expectations-generated inflation are not due to demand excess.
- Even with zero output gap, not all inflation and unemployment can be eliminated.
- Keynesians argue for policies tailored to specific problems, with a focus on managing aggregate demand

# The Phillips Curve

- A.W. Phillips found an inverse relationship between wage inflation and unemployment, known as the Phillips curve.
- This relationship was used to show short-term effects of changes in aggregate demand.
- A rise in demand decreases unemployment and increases inflation; a decrease in demand has the opposite effect.



# The Phillips Curve

- The curve is bowed in because as demand increases, labor becomes scarcer, and wage rates rise.
- The curve shifts with non-demand factors like frictional unemployment and cost-push inflation.
- Policymakers could trade inflation for unemployment, but beyond the short-run, this relationship may not hold.

# The Phillips curve



- ✦ Bowed in to the origin
- ✦ Position depends on non-demand factors
- ✦ Seemed to present government with a simple policy choice

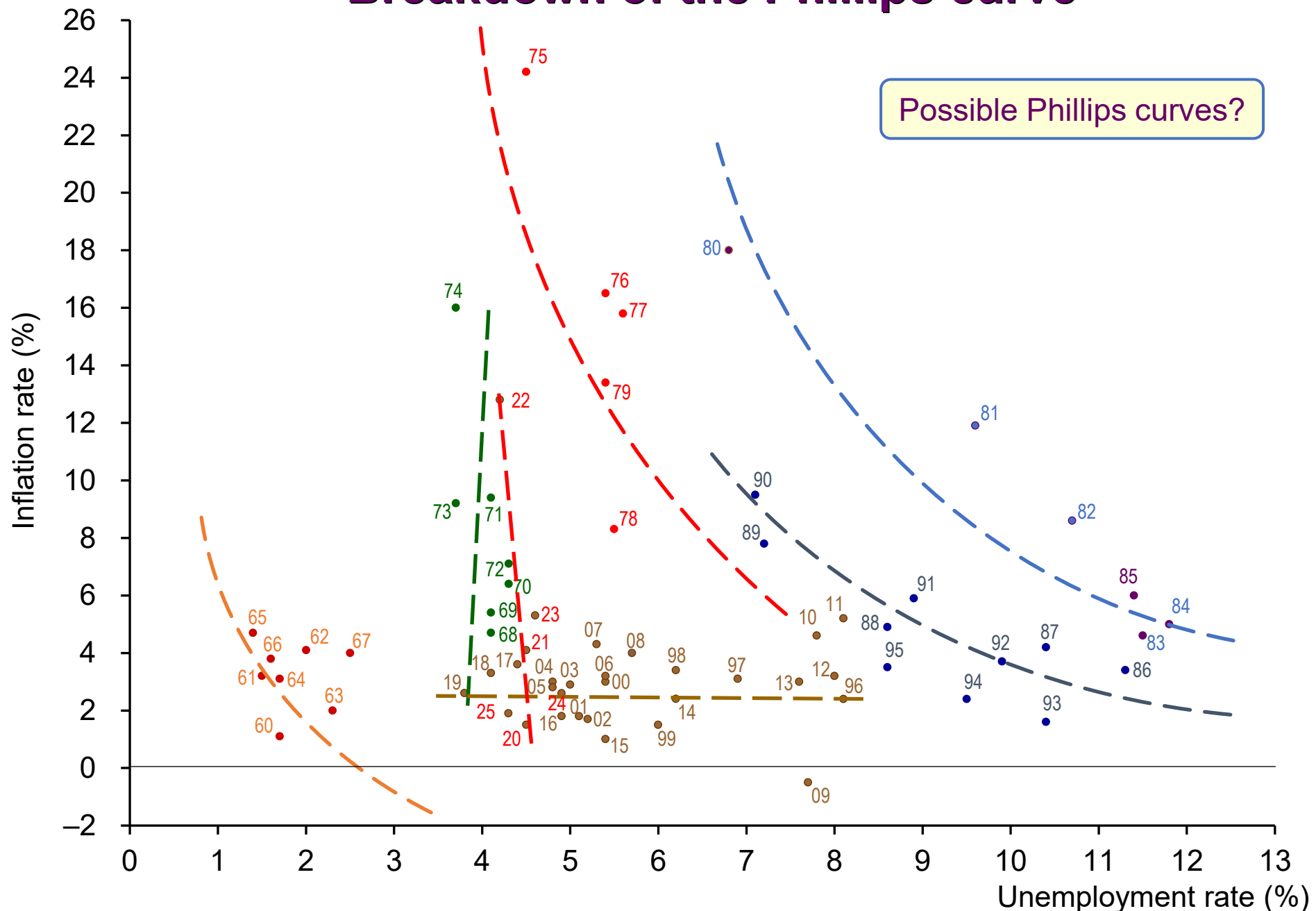
# Breakdown of the Phillips Curve (1967 Onwards)

- From 1967, the relationship between inflation and unemployment began to break down in many countries.
- The UK experienced both high unemployment and high inflation, contradicting the Phillips curve.
- The relationship did not hold consistently from the late 1960s to the early 1990s.

# Breakdown of the Phillips Curve (1967 Onwards)

- Since 1997, the UK has targeted inflation, and the Phillips curve appears to have flattened.
- Inflation rates fluctuated in the late 2000s due to global economic shocks and supply-side factors, similar to the 1970s.

# Breakdown of the Phillips curve



Source: Based on data from the Office for National Statistics; forecasts based on data from *World Economic Outlook Database* (IMF)

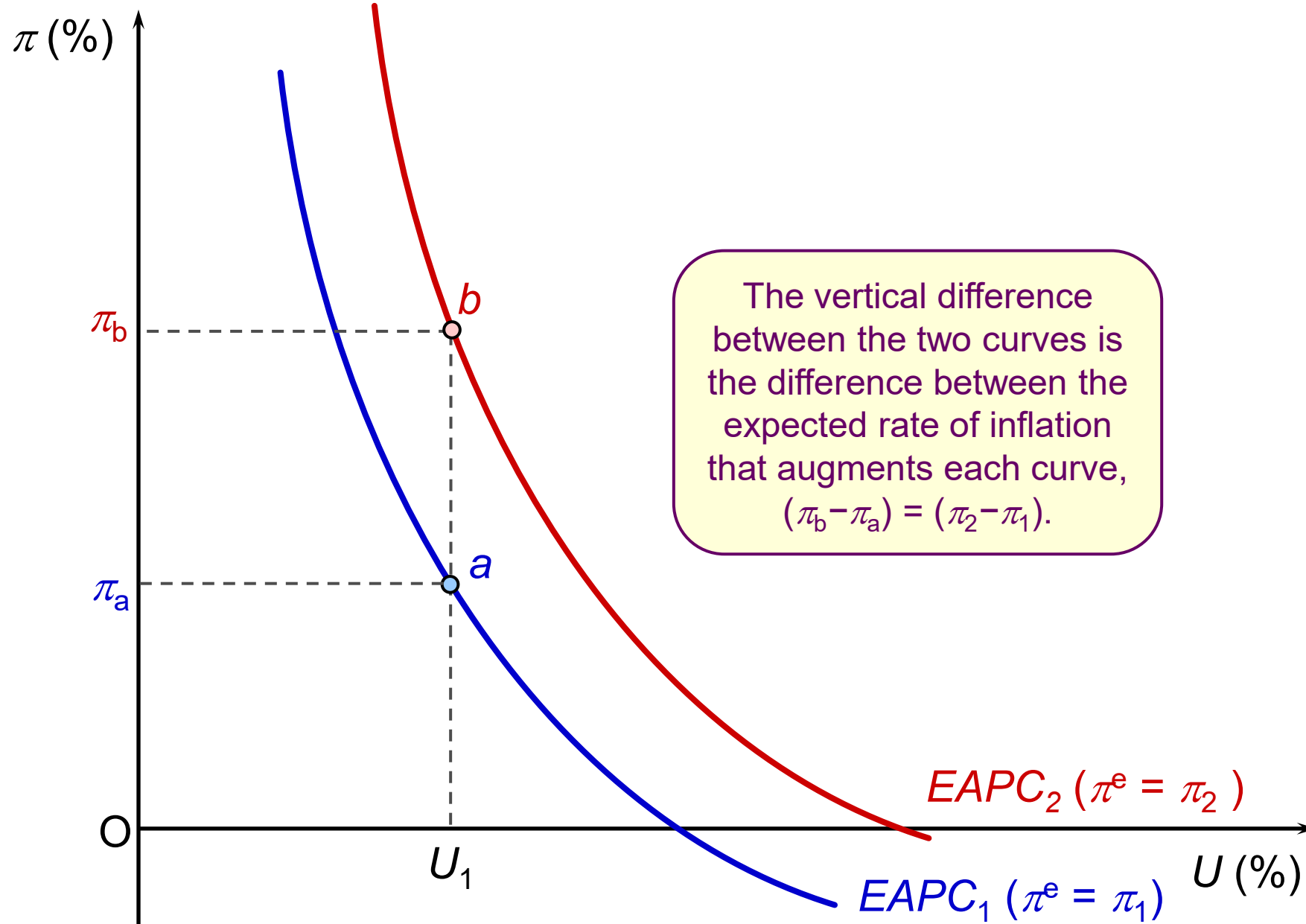
# The Expectations-Augmented Phillips Curve (EAPC)

- The expectations-augmented Phillips curve considers:
  - The inverse of unemployment ( $1/U$ ) for demand-pull inflation.
  - Expected inflation ( $p_e$ ) based on past experiences.
  - Exogenous cost pressures ( $k$ ), like commodity price increases.
- Example: If expected inflation is 3%, excess demand adds 2%, and exogenous costs add 1%, actual inflation would be 6%.
- The expectations-augmented curve forms a family of curves, with each associated with a particular rate of inflationary expectations.

# The Expectations-Augmented Phillips Curve (EAPC)

- Expectations-augmented Phillips curve (*EAPC*)
  - $\pi = f\left(\frac{1}{U}\right) + \pi^e + k$
  - i.e. inflation depends on
    - aggregate demand shown by inverse of unemployment ( $1/U$ )
    - inflation expectations ( $\pi^e$ )
    - cost-pressures ( $k$ )
  - a rise in  $\pi^e$  shifts *EAPC* upwards

# Expectations of inflation and the Phillips curve





# Inflation Targeting and Unemployment

## Inflation targets

| Country      | Inflation target (%) | Details                                                                                            |
|--------------|----------------------|----------------------------------------------------------------------------------------------------|
| Australia    | 2–3                  | Average over the business cycle                                                                    |
| Brazil       | 3.5                  | Tolerance band of $\pm 1.5\%$ points                                                               |
| Canada       | 2                    | Tolerance band of $\pm 1\%$ point                                                                  |
| Chile        | 3                    | Tolerance band of $\pm 1\%$ point                                                                  |
| Czech Rep.   | 2                    | Tolerance band of $\pm 1\%$ point                                                                  |
| Eurozone     | 2                    | Average for eurozone; over medium term                                                             |
| Hungary      | 3                    | Tolerance band of $\pm 1\%$ point; over medium term                                                |
| Iceland      | 2.5                  | Tolerance band of $\pm 1.5\%$ points                                                               |
| Israel       | 1–3                  |                                                                                                    |
| Japan        | 2                    |                                                                                                    |
| Mexico       | 3                    | Tolerance band of $\pm 1\%$ point                                                                  |
| New Zealand  | 2                    | Tolerance band of $\pm 1\%$ point                                                                  |
| Norway       | 2                    | Close to 2% over time                                                                              |
| Peru         | 2                    | Tolerance band of $\pm 1\%$ point                                                                  |
| Poland       | 2.5                  | Tolerance band of $\pm 1\%$ point                                                                  |
| South Africa | 3–6                  |                                                                                                    |
| South Korea  | 2                    |                                                                                                    |
| Sweden       | 2                    | 1–2 year horizon; tolerance band of $\pm 1\%$ point                                                |
| Switzerland  | <2                   | Close to 2%                                                                                        |
| Thailand     | 1–3                  | Tolerance band of $\pm 1.5\%$ points                                                               |
| UK           | 2                    | Forward looking; tolerance band of $\pm 1\%$ point                                                 |
| USA          | 2                    | Average rate of 2% over time; consistent with dual mandate of stable prices and maximum employment |

Source: Various central bank websites (see [BIS Central bank hub](#); see also [Central Bank News](#))

# Inflation Targeting and Unemployment



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- Inflation targeting is central to macroeconomic policy for many countries.
- Inflation targets are generally achievable, but high interest rates to reduce inflation may cause recessions
- .Once inflation is controlled, countries tend to maintain it at the target level, leading to expectations of sustained success.

# Inflation Targeting and Unemployment



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## Problem with inflation targeting:

- Phillips curve has become nearly horizontal, showing less correlation between inflation and business cycles.
- Under inflation targeting, booms may not generate inflation, leading to unsustainable growth followed by recessions.
- Changes in equilibrium unemployment can shift the Phillips curve, affecting inflation and unemployment dynamics.

# Inflation Targeting and Unemployment



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## Financial cycles:

- Low interest rates in the 2000s led to unsustainable lending, increasing financial instability and asset prices.
- A balance sheet recession occurred when excessive investments led to financial distress.

# Inflation Targeting and Unemployment

## Cost-push inflation dilemma:

- If inflation is driven by costs, interest rate hikes to control inflation may worsen output gaps.
- Examples: 2010-2012 post-crisis period and inflationary shocks from 2021-2022 due to rising commodity prices.

# The Volatility of private-sector Spending and the Business Cycle

# The Composition of Aggregate Demand

- Aggregate demand is key to short-term economic activity; rapid growth is linked to expanding demand, while recessions to declining demand.
- Understanding what drives aggregate demand is crucial for understanding the business cycle

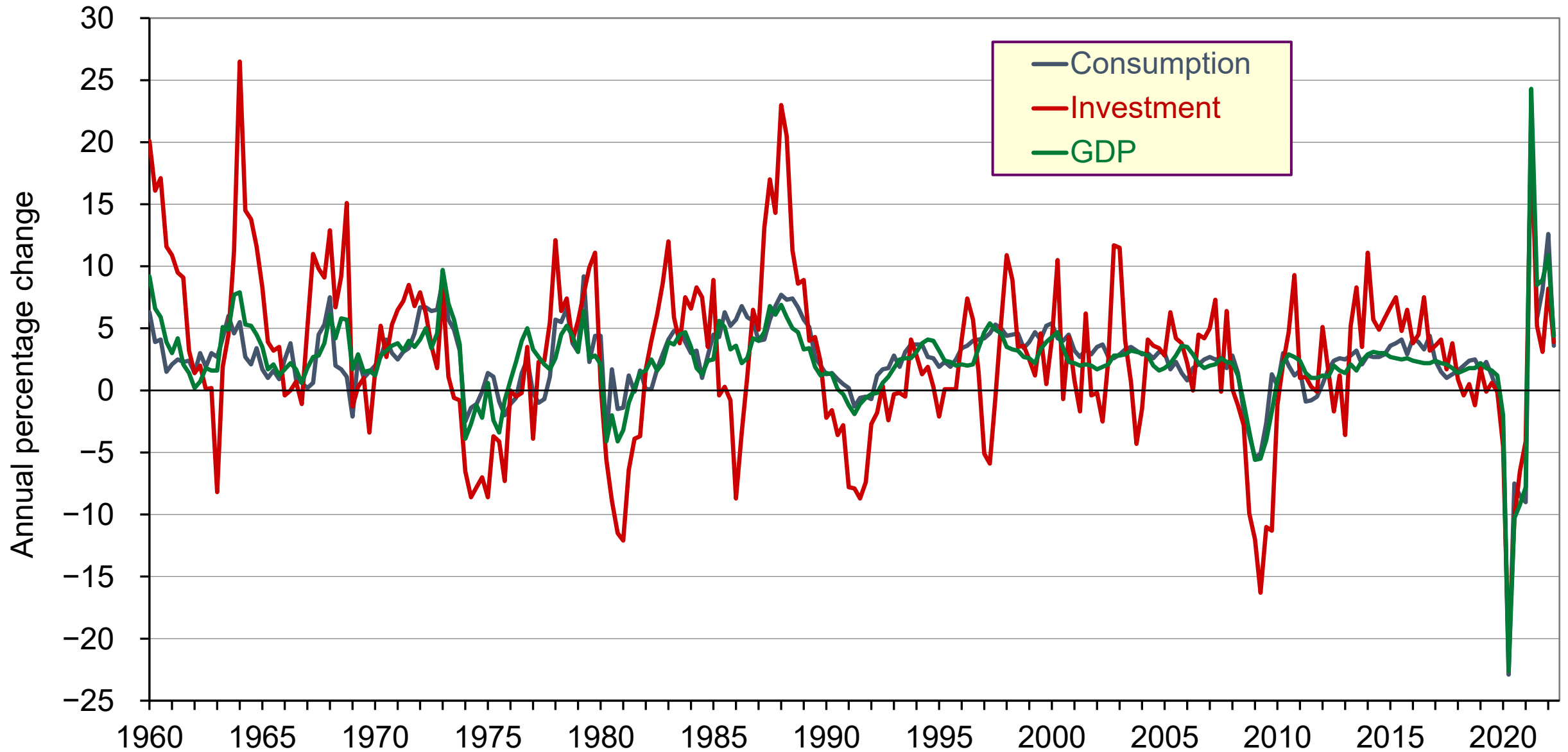


# Composition of aggregate demand, % (average 1970–2020)

|             | Household<br>final<br>consumption | Gross capital<br>formation<br>(public and<br>private) | Government<br>final<br>consumption | Exports | Imports | External<br>balance<br>( $X - M$ ) |
|-------------|-----------------------------------|-------------------------------------------------------|------------------------------------|---------|---------|------------------------------------|
| Australia   | 56.5                              | 26.6                                                  | 17.8                               | 17.8    | 18.4    | -0.7                               |
| Brazil      | 64.8                              | 19.8                                                  | 15.8                               | 10.4    | 10.4    | 0.0                                |
| China       | 45.1                              | 38.9                                                  | 14.2                               | 16.9    | 15.2    | 1.7                                |
| France      | 54.7                              | 23.2                                                  | 22.1                               | 24.1    | 24.1    | 0.0                                |
| Germany     | 56.0                              | 23.5                                                  | 19.6                               | 29.0    | 27.5    | 1.5                                |
| India       | 66.7                              | 27.1                                                  | 10.5                               | 12.2    | 14.1    | -1.9                               |
| Ireland     | 52.7                              | 24.7                                                  | 17.5                               | 72.7    | 66.6    | 6.1                                |
| Japan       | 53.1                              | 29.9                                                  | 16.1                               | 12.7    | 11.8    | 0.9                                |
| Netherlands | 49.1                              | 22.3                                                  | 22.7                               | 60.5    | 54.4    | 6.1                                |
| Singapore   | 45.0                              | 33.3                                                  | 10.2                               | 175.5   | 163.9   | 11.5                               |
| Sweden      | 47.8                              | 24.9                                                  | 25.2                               | 35.7    | 32.8    | 2.9                                |
| UK          | 66.2                              | 20.1                                                  | 19.5                               | 26.0    | 26.8    | -0.9                               |
| USA         | 64.6                              | 22.0                                                  | 15.6                               | 9.7     | 11.9    | -2.2                               |

Source: *National Accounts Main Aggregates Database* (United Nations Statistics Division)

# Annual growth of UK consumption, investment and output



Notes: Annual growth rates are calculated using constant-price data; investment growth is growth in Gross Fixed Capital Formation (whole economy).

Source: Based on series [KGZZ](#), [KG7T](#) and [IHYR](#) (ONS, 2022)

# The Volatility of Private-sector Spending

- Business cycles and the composition of aggregate demand (cont.)
  - instability of investment
    - the investment accelerator
      - induced investment:  $I_t = \alpha \Delta Y$  ( $\alpha$  = accelerator coefficient)
    - determinants of the size of the accelerator
      - size of  $\alpha$  depends on marginal capital-output ratio ( $\Delta K / \Delta Y$ )

# The Investment Accelerator:

- Investment rises rapidly when an economy recovers from a recession and falls dramatically during a downturn.
- Investment is an injection into the circular flow of income, causing multiplied changes in national income that amplify booms or deepen recessions.

# The Investment Accelerator:

- **Accelerator theory:** A modest rise in national income leads to a much larger percentage rise in investment.
- **Induced investment:** New investment to increase production capacity when income and consumption rise.
- **Replacement investment:** Investment needed to replace outdated or worn-out capital, occurring when no changes in income occur.
- The accelerator coefficient ( $a$ ) determines the extent to which investment responds to changes in national income.

# The Investment Accelerator:

## Key Factors:

- Marginal capital-output ratio ( $\Delta K/\Delta Y$ ): The ratio of capital investment required to produce additional output, affecting the accelerator coefficient.
- Firms may have spare capacity or carry stocks, delaying investment decisions.
- The accelerator effect exists, though difficult to predict, as firms' investment spending responds to changes in aggregate demand.

# The Interaction of the Multiplier and Accelerator:

- The multiplier effect amplifies the accelerator effect, with increased government purchases leading to a rise in income and induced investment, further boosting income.
- Explosion of national income?: No, because:
  - National income cannot grow faster than potential output (limited by full employment).
  - To sustain investment, national income must rise faster and faster; if growth slows, investment falls, reversing the cycle.

# Other Determinants of Investment:

- **Expectations of Future Market Conditions:** Firms base investment on expectations of future conditions, which can cause bandwagon effects (confidence leading to more investment in expansions).
- **Uncertainty:** High uncertainty about future conditions makes firms more cautious, reducing investment.
- **Business Confidence:** Sentiment about market conditions significantly affects investment decisions; pessimism can drastically reduce investment.



# Other Determinants of Investment:

- **Availability of Finance:** Investment often requires financing; difficulties in raising finance (e.g., during financial crises) can limit investment.
- **Rate of Interest:** High interest rates make investment more expensive, reducing profitability and investment levels.
- **Cost and Efficiency of Capital Equipment:** Lower capital costs or more efficient machines increase investment returns, encouraging firms to invest more. Technological progress plays a key role.

# The Financial Sector:

- The financial sector can dampen or create volatility in spending:
  - Dampens cycles through short-term credit, helping businesses and people manage their spending.
  - Creates volatility due to changes in credit availability or costs and the impact on sector balance sheets.

# The Financial Sector:

## Source of Volatility:

- Financial institutions can cause volatility through excessive lending and asset purchases, leading to unsustainable growth and eventual downturns.
- Balance sheet recession: Financially distressed sectors burdened by debt, as seen during the 2007-2009 financial crisis, contribute to economic downturns.
- Financial distress is an early indicator of potential crises.

# The Financial Sector:

## **Financial Sector as a Magnifier of Volatility:**

- Financial institutions amplify economic shocks by adjusting lending practices.
- Financial accelerator effect: Rising income encourages easier credit, increasing aggregate demand; falling income leads to tighter credit and decreased demand.

# Turning Points in Economic Cycles:

- Ceilings and Floors:
  - Economic growth slows once full capacity is reached (ceiling to output). During recessions, a minimum consumption level remains (floor to consumption).
- Echo Effects:
  - Replacing durable goods and capital purchased in previous booms helps end recessions.
- Accelerator:
  - Investment growth requires continually rising consumer demand; if demand slows, investment falls, and the boom ends.

# Turning Points in Economic Cycles:

- Sentiment and Expectations:
  - Changes in sentiment can influence expectations of boom or recession duration.
- Random Shocks:
  - External political, social, or natural events affect economic sentiment and aggregate demand.
- Government Policy Changes:
  - Governments may use contractionary policies in booms (to address inflation and deficits) and expansionary policies in recessions (to tackle unemployment and stagnation).
  - Successful policies can alter the course of economic cycles, as seen in responses to the global recession of the late 2000s.

# Questions